

COMSATS UNIVERSITY ISLAMABAD, LAHORE CAMPUS

Course Title:	Machine Learning Fundamentals	Course Code:	AIC354
Credit Hours:	3(2,1)	Program Batch:	BS (CS)
Course Instructor:	Dr. Rao Muhammad Adeel Nawab	Semester:	Fall 2026
Submission Deadline:	Sunday, October 11, 2026 at 23:59	Maximum Marks:	40

Assignment 2: Developing an End-to-End Machine Learning System

ASSIGNMENT AIM

Learn how to build a complete Machine Learning system from data collection to deployment using Structured Data.

LEARNING OUTCOMES

By completing this assignment, you will learn how to:

- Handle Data Collection, Data Understanding, Data Preprocessing, and Label Encoding
- Execute the four phases of the Machine Learning Cycle
- Deploy a trained Machine Learning model on the cloud

OVERVIEW & APPROACH

This assignment uses a Half-Cooked Approach:

- Some work has already been completed by the instructor.
- You are provided with the full code for a Titanic Passenger Survival Prediction System.
- You are provided with a tutorial on how to deploy a model to the cloud using PythonAnywhere.

Note: Complete all work using a Jupyter Notebook.

ASSIGNMENT TASKS

Task 1: Titanic Passenger Survival Prediction System

Run & Understand: Execute and review the provided Titanic Passenger Survival Prediction System code on your local machine.

Deploy: Deploy the trained Titanic Passenger Survival Prediction model to the cloud (e.g., PythonAnywhere).

Task 2: Heart Disease Prediction System

Download Data: Download the Heart Disease Prediction Dataset from UCI:

- Dataset Link: <https://archive.ics.uci.edu/dataset/45/heart+disease>

Build System: Modify the Titanic Passenger Survival Prediction code to build a Heart Disease Prediction System following the exact same steps.

- Constraint: Select only 4 input attributes from the Heart Disease dataset.

Deploy: Deploy your trained Heart Disease Prediction model to the cloud.

GITHUB REPO AND LIVE DEMO

The code and dataset are available on GitHub repo on the following link.

Repo: [\[https://github.com/haqnawaz99/ml-teaching-notebooks\]](https://github.com/haqnawaz99/ml-teaching-notebooks)

The ML Model is deployed and made live on the following link.

Live demo: [\[https://ml-teaching-notebooks-jrrdwerctxskw5kwc6rufvp.streamlit.app/\]](https://ml-teaching-notebooks-jrrdwerctxskw5kwc6rufvp.streamlit.app/)

SUBMISSION GUIDELINES

- **Platform:** Upload your code, dataset, and trained model file to Google Classroom.
- **Deployment Link:** Include your live cloud deployment link at the very top of your Jupyter Notebook file.

POLICIES

Penalty for Late Submission

- **Deduction:** 5% mark reduction per day for late submissions.
- **Cutoff:** Submissions made 3 days after the deadline will not be accepted (0 Marks).

Plagiarism Policy

Institutional policy regarding punishment will be strictly enforced if a student is found guilty of plagiarism.